Design of Experiments (DoE)

Modelització Estadística

Grau en Intel·ligència Artificial

Course 2024-25

Outline

Introduction to Design of Experiments (DOE)

What is DOE?

Design of Experiments (**DOE**) is a structured methodology for planning and conducting experiments efficiently:

- **Objective**: Identify the factors that **influence an outcome** and optimize performance.
- **Why Use DOE?**:
  - Reduces experimentation time and cost.
  - Provides insights into interactions between variables.
  - Ensures reliable conclusions by controlling for randomness.

Why Experiment?

- **Causality**: Experiments reveal cause-and-effect relationships, unlike observational studies.
- **Control**: Allows systematic manipulation of factors to isolate their effects.
- **Insights**: Helps prioritize variables with the highest impact on outcomes.

Example

Example

We want to improve our knowledge in a machine learning process:

- **Feature Engineering**: Test the impact of different feature transformations on model performance.
- **Algorithm Tuning**: Compare algorithms under controlled conditions to select the best one for a task.
- **Hyperparameter Optimization**: Identify the best combinations of parameters (e.g., learning rate, batch size).

Factors to test:

- **Preprocessing**: Standardization vs. normalization.
- **Algorithm**: Logistic regression vs. random forest.
- **Hyperparameters**: Learning rate (low vs. medium vs. high).

A DOE approach assesses these combinations efficiently, revealing the best setup.

Factors and Levels

What are Factors and Levels?

- **Factors**: Variables that can influence the outcome of an experiment.
  - Examples in AI: Learning rate, number of layers, optimizer type.
- **Levels**: The specific values or settings that a factor can take.
  - Examples: Low, Medium, High for learning rate.

Types of Factors

- **Categorical Factors**:
  - Discrete options or categories.
  - Example: Optimizer type (`Adam`, `SGD`, `RMSprop`).
- **Continuous Factors**:
  - Numeric variables with a range of possible values.
  - Example: Learning rate $(0.001, 0.01, 0.1)$.

Remark

A careful **selection of factors and levels** is crucial for meaningful experimentation.

Example

Context: AI Pipeline Optimization

- Factors:
  - **Data Augmentation**: Categorical (`Yes`, `No`).
  - **Batch Size**: Continuous $(16, 32, 64)$.
- Levels:
  - **Data Augmentation**:
    - Level 1: `Yes`.
    - Level 2: `No`.
  - **Batch Size**:
    - Level 1: 16.
    - Level 2: 32.
    - Level 3: 64.
- Combining these factors and levels creates an experimental design.

Outcome

Outcome

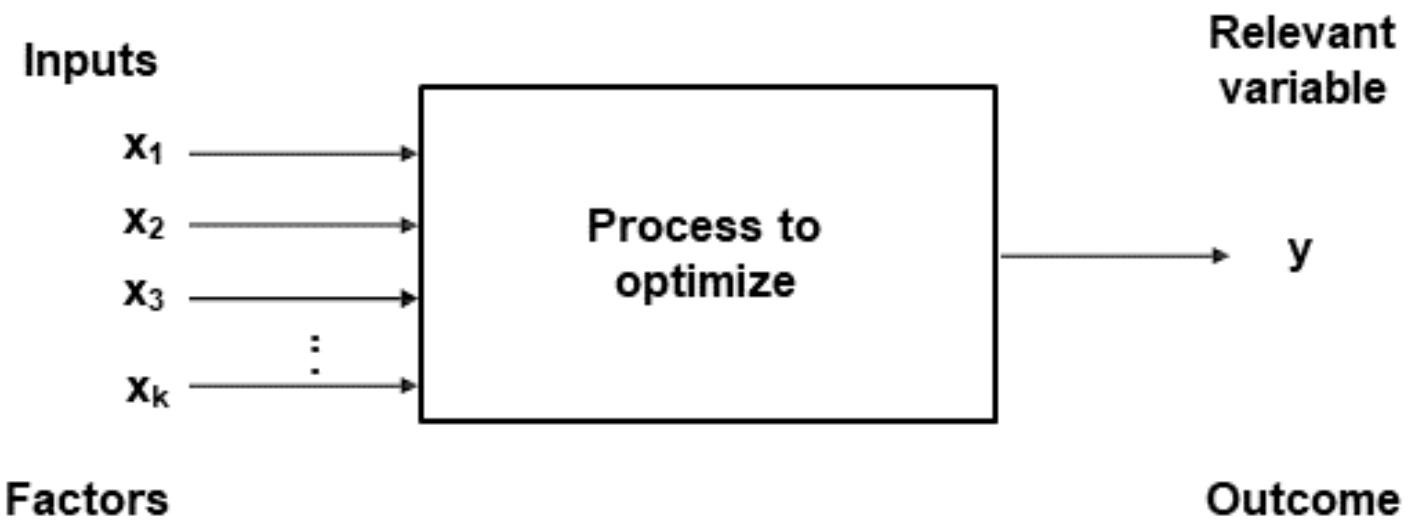
In DOE, the **outcome** is the variable to optimize during the experiment:

- Represent the results of **interest** influenced by the factors.
- Outcomes are analyzed to understand the **effects** of factors and their **interactions**.

Example

- **Problem**: Optimizing a neural network model.
- **Outcomes**:
  - **Accuracy**: Percentage of correct predictions on the test set.
  - **Training Time**: Total time taken to complete training.
- **Goal**: Maximize accuracy constrained to a fixed training time.

General schema



Simulations. Pros and cons

- An alternative to perform experiments is to simulate data.

Pros

Simulations can reduce the standard error to 0 (not the standard deviation).

- Despite the usefulness of simulations, some situations require experimentation.

Cons

The main challenge lies in systematic errors or inaccuracies in model specification.

- We **lack the knowledge** to create accurate models or face situations that are complex to model.
- We **do not have empirical data** to estimate certain parameters.
- The models we assume to be true **are not accurate** representations of reality.

The Need of Experimentation

While experimentation is necessary for addressing uncertainties and validating models, it comes with challenges:

- **Experimentation is expensive**: Both in terms of resources and time.
- **Balancing Simulations and Experiments**:
  - Use simulations to refine hypotheses and reduce the experimental space.
  - Combine both approaches to achieve reliable and cost-effective results.

Interactions

What are Interactions?

- **Interactions** occur when the effect of one factor depends on the level of another factor.
- They reveal how factors work together rather than independently.

Example

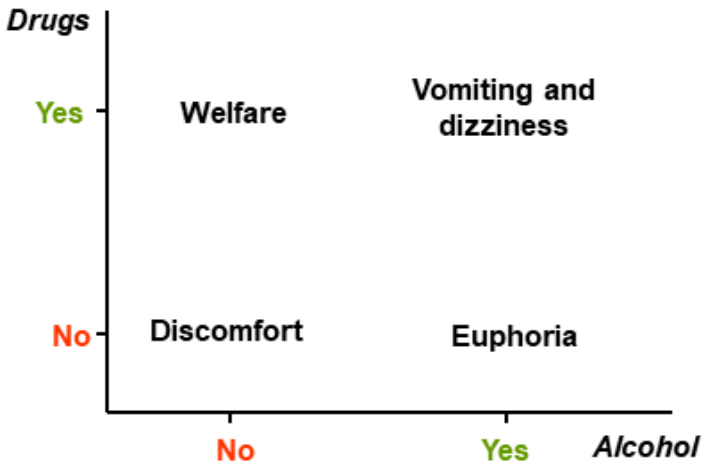
The impact of **learning rate** on model accuracy might depend on the **batch size**.

Why Are Interactions Important?

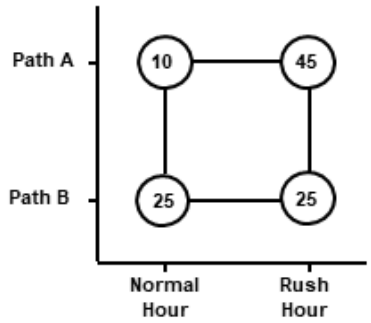
- **Understanding interactions** helps identify effects between factors.
- Ignoring interactions can lead to **misleading conclusions**.
- Interaction effects can guide **fine-tuning of AI pipelines**.

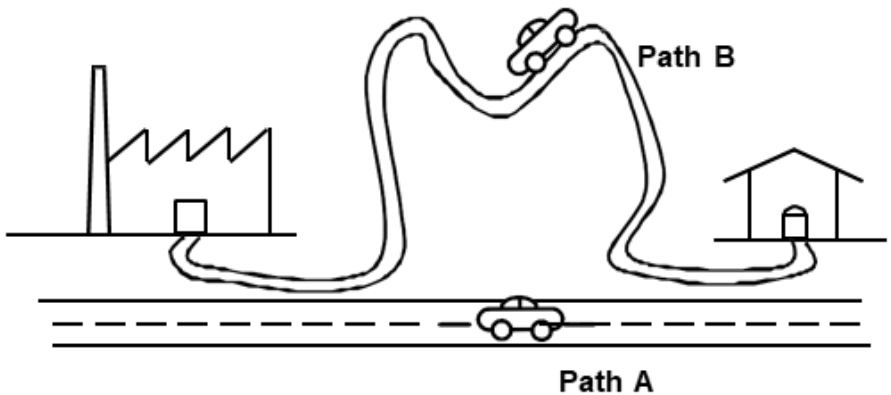
Interactions. Examples

Drugs
vs.
Alcohol



Path
vs.
Time slot





Synergistic and Antagonistic Factors

Synergistic Factors

Two factors are **synergistic** when the effect of applying them together is **greater** than the sum of their individual effects.

- An increase in pressure alone results in a gain of $+5$ in the product, and an increase in temperature alone results in a gain of $+5$. An increase in both results in a gain of $+15$.

Antagonistic Factors

Two factors are **antagonistic** when the effect of applying them together is **less** than the sum of their individual effects.

- An increase in pressure alone results in a gain of $+5$ in the product, and an increase in temperature alone results in a gain of $+5$. However, an increase in both results in a gain of $+8$.

Remark

To discuss synergistic or antagonistic factors, **reference levels** must be established, such as the *absence of...* a condition.

Visualize interactions

Example

- **Problem**: Evaluate the combined effect of `learning rate` and `batch size` on model accuracy.
- **Visualization**: `interaction.plot` in R
- **Interpretation**:
  - **Parallel lines** indicate no interaction.
  - **Diverging lines** suggest strong interaction effects.

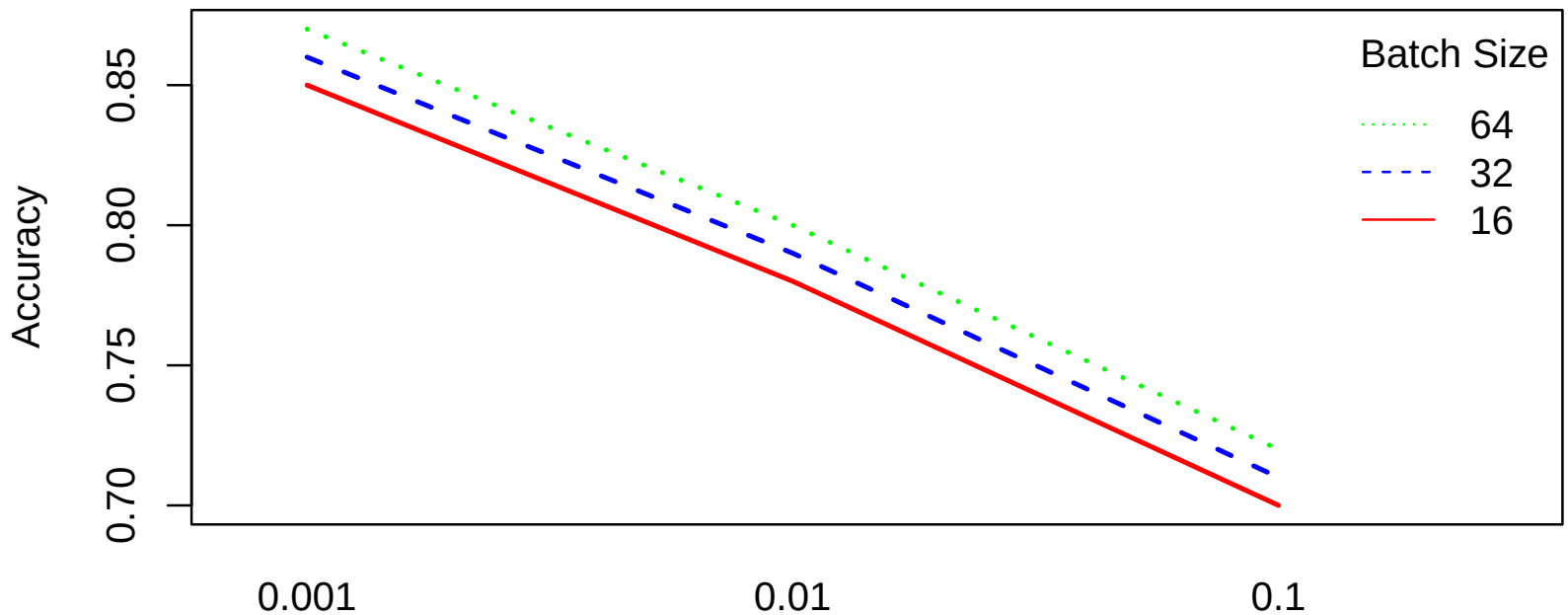
Visualize interactions. Data without interaction

- **Factors**: Learning rate, Batch size
- **Outcome**: Accuracy

```
##   learning_rate batch_size accuracy
## 1         0.001         16     0.85
## 2         0.001         32     0.86
## 3         0.001         64     0.87
## 4         0.010         16     0.78
## 5         0.010         32     0.79
## 6         0.010         64     0.80
## 7         0.100         16     0.70
## 8         0.100         32     0.71
## 9         0.100         64     0.72
```

Visualize interactions. Interaction Plot (NO interaction)

```
interaction.plot(x.factor     = data$learning_rate,
                 trace.factor = data$batch_size,
                 response     = data$accuracy,
                 xlab = "Learning Rate", ylab = "Accuracy",
                 trace.label = "Batch Size", lwd=2, lty = 1:3,
                 col = c("red", "blue", "green"),  pch = 16:18)
```



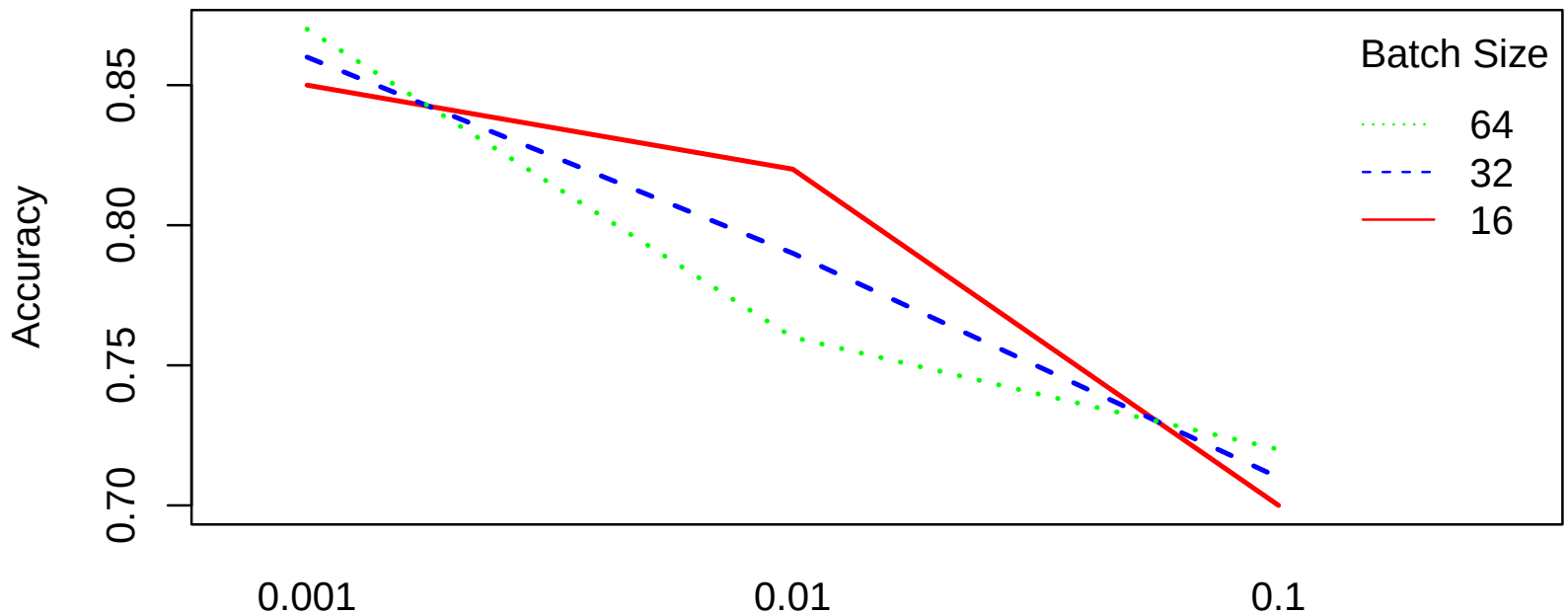
Visualize interactions. Data with interaction

- **Factors**: Learning rate, Batch size
- **Outcome**: Accuracy

```
##   learning_rate batch_size accuracy
## 1         0.001         16     0.85
## 2         0.001         32     0.86
## 3         0.001         64     0.87
## 4         0.010         16     0.78
## 5         0.010         32     0.79
## 6         0.010         64     0.80
## 7         0.100         16     0.70
## 8         0.100         32     0.71
## 9         0.100         64     0.72
```

Visualize interactions. Interaction Plot (Interaction)

```
interaction.plot(x.factor     = data2$learning_rate,
                 trace.factor = data2$batch_size,
                 response     = data2$accuracy,
                 xlab = "Learning Rate", ylab = "Accuracy",
                 trace.label = "Batch Size", lwd=2, lty = 1:3,
                 col = c("red", "blue", "green"), pch = 16:18)
```



Strategies. Using existing data

Alternative to Experimentation: Old data

The first alternative to save on experimentation costs could be to use already existing data.

- If the data are **observational** (not from an experiment), we cannot establish causality.
  - **Example:** Did the process deteriorate in February because we changed Machine A, or because Employee B left?
- If the data come from an **experiment**, the main issue is that the data might be inconsistent due to changes over time:
  - **Aging:** Components or materials may degrade over time.
  - **Repairs:** Equipment adjustments can introduce variability.
  - **Procedure Changes:** Modifications in the process may affect results.

Strategies. Plan

Experimenting Without a Plan

Making changes to the system and observing how productivity changes.

- This approach is **not recommended.**

Designing the Entire Plan Upfront

- If we know the system well, this can be an alternative.
- Otherwise, it is not advisable to invest the entire budget in a single experiment.
  - There are many things that could go wrong!

Sequential Strategy

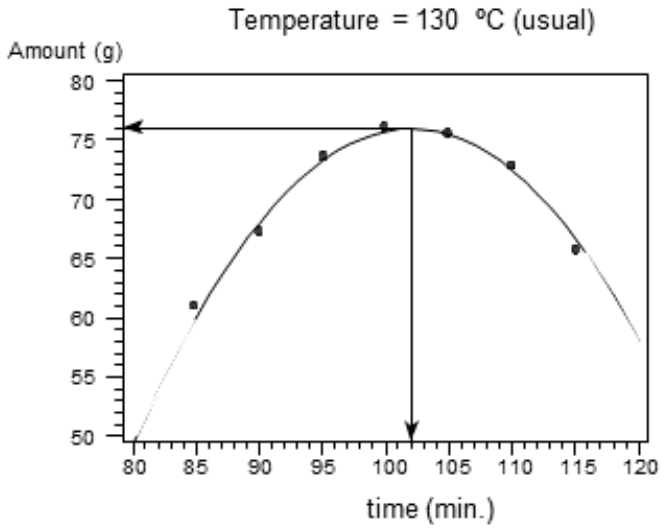
This is the **preferred strategy.**

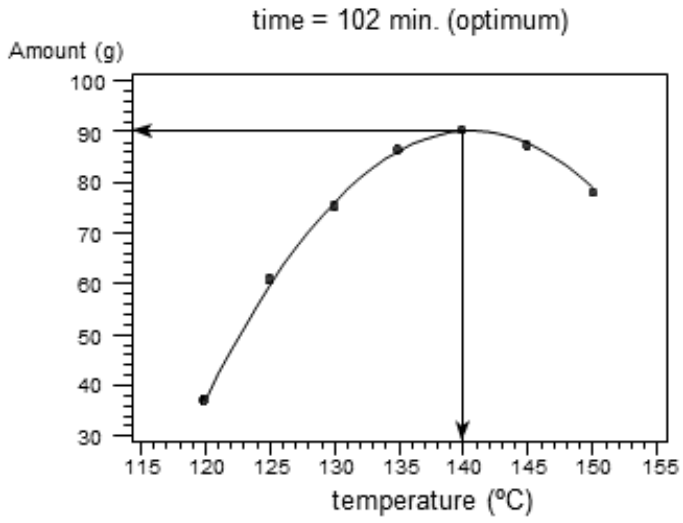
- A first phase can be carried out with real experimentation or simulation.
- A second phase can follow with real experimentation.

Strategies. Naive strategy

Naive strategy

1. It consists of fixing the value of all factors except one, which will be varied.
2. Find the optimal value of the unfixed factor and assign it that value.
3. Repeat the same process with the remaining factors.



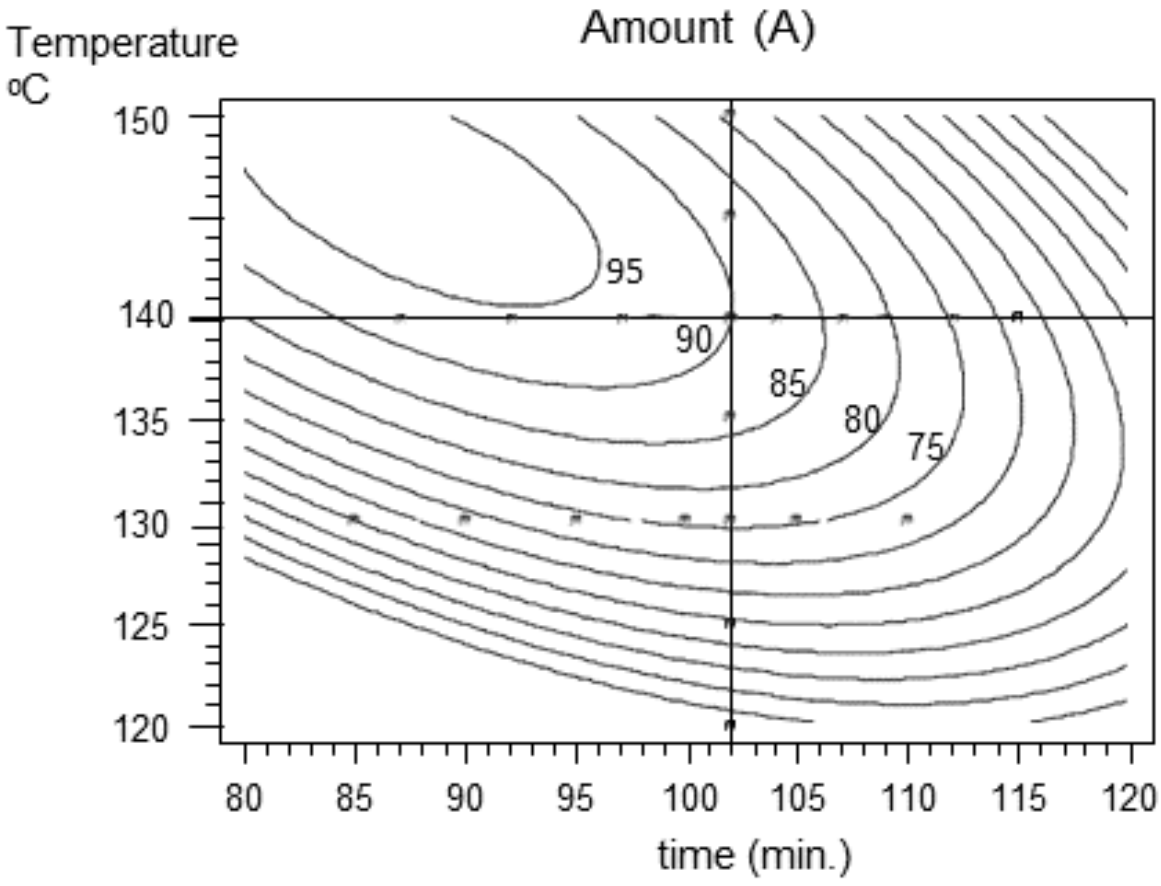


Optimum:

T = 140 ºC ; t = 102 min → A = 90 g

¿Correct?

Strategies. Naive strategy. Problem



Problem of Naive Strategy

If there are interactions, the **naive strategy does not properly work**

Outline: Types of designs

1. Full Factorial Design (FFD)
2. Fractional Factorial Designs (FD)
3. Latin Squares (LS)
4. Optimal Designs

Full Factorial Design

What is a Full Factorial Design?

A **full factorial design** (FFD) is an experimental setup where all possible combinations of factor levels are tested:

- **Objective**: Study the effect of each factor and their interactions on the outcome.
- **Structure**:
  - k factors with l_i levels ($i = 1, 2, \ldots, k$).
  - Total combinations: $\prod_{i=1}^{k} l_i$.
- Suitable for understanding:
  - **Main effects** of factors.
  - **All** interactions between factors.

Example

- Factors:
  - **Learning Rate** ($I_1 = 2$): `0.01`, `0.1`.
  - **Batch Size** ($I_2 = 3$): `16`, `32`, `64`.
- Total combinations: $2 \times 3 = 6$.
- Experimental layout:

Learning Rate	Batch Size
0.01	16
0.01	32
0.01	64
0.1	16
0.1	32
0.1	64

Pros of FFD

- **Thorough Analysis**:
  - Identifies main effects and all interaction effects.
- **Accurate Insights**:
  - Ensures no important effects are overlooked.
- **Applicable to Small Designs**:
  - Useful when the number of factors and levels is manageable.

Cons of FFD

- **Exponential Growth of Combinations**:
  - Adding factors or levels significantly increases the total runs.
- **Resource Intensive**:
  - Time and computational costs grow quickly with the number of combinations.
- **Cost**:
  - Expensive in real situations.
- **Complex Analysis for Many Factors**:
  - High-dimensional data can be challenging to interpret.

R

Use the DoE.base package to generate a full factorial design.

```r
library(DoE.base)
# Define factors and levels
factors <- list(learning_rate = c(0.01, 0.1),
                batch_size    = c(16, 32, 64))
# Generate design
full_design <- fac.design(nlevels=c(2,3),nfactors = 2,
                          factor.names = names(factors))
full_design # Combinations are intentionally unsorted
```

```
##   learning_rate batch_size
## 1             1          2
## 2             2          3
## 3             2          1
## 4             2          2
## 5             1          3
## 6             1          1
## class=design, type= full factorial
```

Design matrix

What is a Design Matrix?

- A **design matrix** is a structured representation of all the experimental runs in a factorial experiment.
- It shows the levels of each factor for every experimental combination.
- In the context of a two-level factorial design, the levels are coded as:
  - $+1$: Represents the high or maximum level of a factor.
  - -1: Represents the low or minimum level of a factor.

Benefits of Using Coded Levels (-1 and +1)

- Simplifies calculations for estimating effects and interactions.
- Ensures all factors are on the same scale, avoiding unit mismatch.
- Enables easier interpretation of factor effects.

Example Design Matrix for Two Factors

Run	Factor A	Factor B
1	-1	-1
2	-1	$+1$
3	$+1$	-1
4	$+1$	$+1$

Fractional Factorial Design

What is a Fractional Factorial Design?

A **fractional factorial design** is a reduced version of a full factorial design:

- **Objective**: Test a carefully selected subset of all possible factor-level combinations.
- **Why Use It?**
  - Reduces the number of experiments while retaining critical information.
  - Useful when resources are limited, but key effects still need to be estimated.

Key Features of FD Construction

- The goal is to estimate the **maximum number of effects** while omitting unnecessary combinations.
- Inevitably, some effects will be **confounded** (cannot be distinguished from one another).
- Main effects should ideally be confounded with interactions of the **highest order possible**.

Ortogonality

What is Orthogonality?

- Orthogonality refers to the relationship between two vectors in a vector space. Two vectors are said to be orthogonal if their **dot product is zero**.
- A fractional factorial design (FD) omits some combinations of factor levels while attempting to maintain **orthogonality**.
- Orthogonality **ensures independence among the estimated effects**, enabling clear interpretation of the results:
  - The estimated coefficients are the same regardless the factors included in the model.

Fractional Factorial Designs construction. Generators

Generators

- To construct a FD from a FFD, a **generator** is needed.
- The choice of the generator considers the **principles of orthogonality and confounding**.
  - **Ortogonalithy** is guarantee in a FD.
  - A FD ensures main effects are **confounded** with the highest-order interactions possible.
- A generator is a **product of columns** of the design matrix that serves to generate a new column.
- In general, how to choose a generator **is not trivial**.
- For instance, for constructing a fractional factorial design (2^4) from a full factorial design (2^5):
  - A full factorial design for 4 factors (A, B, C, D) is constructed first.
  - Factor E is defined using the generator $E = ABCD$, where
  - The value of E is calculated by multiplying the levels of A, B, C, and D (e.g., $E = (+1)(-1)(+1)(+1) = -1$).

Example

- We need 1 generator to go from 16 to 8 experiments.

Prod.	A	B	C	D	E
1	-1	-1	-1	-1	-1
2	1	-1	-1	-1	-1
3	-1	1	-1	-1	-1
4	1	1	-1	-1	-1
5	-1	-1	1	-1	-1
6	1	-1	1	-1	-1
7	-1	1	1	-1	-1
8	1	1	1	-1	-1
9	-1	-1	-1	1	-1
10	1	-1	-1	1	-1
11	-1	1	-1	1	-1
12	1	1	-1	1	-1
13	-1	-1	1	1	-1
14	1	-1	1	1	-1
15	-1	1	1	1	-1
16	1	1	1	1	-1
17	-1	-1	-1	-1	1
18	1	-1	-1	-1	1
19	-1	1	-1	-1	1
20	1	1	-1	-1	1
21	-1	-1	1	-1	1
22	1	-1	1	-1	1
23	-1	1	1	-1	1
24	1	1	1	-1	1
25	-1	-1	-1	1	1
26	1	-1	-1	1	1
27	-1	1	-1	1	1
28	1	1	-1	1	1
29	-1	-1	1	1	1
30	1	-1	1	1	1
31	-1	1	1	1	1
32	1	1	1	1	1

Exp.	A	B	C	D	E
1	-1	-1	-1	-1	1
2	1	-1	-1	-1	-1
3	-1	1	-1	-1	-1
4	1	1	-1	-1	1
5	-1	-1	1	-1	-1
6	1	-1	1	-1	1
7	-1	1	1	-1	1
8	1	1	1	-1	-1
9	-1	-1	-1	1	-1
10	1	-1	-1	1	1
11	-1	1	-1	1	1
12	1	1	-1	1	-1
13	-1	-1	1	1	1
14	1	-1	1	1	-1
15	-1	1	1	1	-1
16	1	1	1	1	1

Confounding Structure

- Main effects (A, B, C, D, E) are confounded with **4-factor interactions** (e.g., $E = ABCD$).
- Two-factor interactions are confounded with **other 3-factor interactions** (e.g., $AE = BCD$).
- Confounding ensures that higher-order interactions (usually negligible) **do not distort** main effect estimates.
- Significant interactions between three or more factors are **rare**.
- Most models are sufficient when they consider:
  - Main effects.
  - Two-factor interactions.

Resolution of the Design

- The **resolution** of a Design is the sum of the order of effects confounded with one another:
  - Main effects confounded with 4-factor interactions: $4 + 1 = 5$.
  - Two-factor interactions confounded with 3-factor interactions: $3 + 2 = 5$.
- Hence, this is a **Resolution V** design.

Example

- We need 2 generators to go from 16 to 4 experiments. **Resolution is III**.

Prod.	A	B	C	D	E
1	-1	-1	-1	-1	-1
2	1	-1	-1	-1	-1
3	-1	1	-1	-1	-1
4	1	1	-1	-1	-1
5	-1	-1	1	-1	-1
6	1	-1	1	-1	-1
7	-1	1	1	-1	-1
8	1	1	1	-1	-1
9	-1	-1	-1	1	-1
10	1	-1	-1	1	-1
11	-1	1	-1	1	-1
12	1	1	-1	1	-1
13	-1	-1	1	1	-1
14	1	-1	1	1	-1
15	-1	1	1	1	-1
16	1	1	1	1	-1
17	-1	-1	-1	-1	1
18	1	-1	-1	-1	1
19	-1	1	-1	-1	1
20	1	1	-1	-1	1
21	-1	-1	1	-1	1
22	1	-1	1	-1	1
23	-1	1	1	-1	1
24	1	1	1	-1	1
25	-1	-1	-1	1	1
26	1	-1	-1	1	1
27	-1	1	-1	1	1
28	1	1	-1	1	1
29	-1	-1	1	1	1
30	1	-1	1	1	1
31	-1	1	1	1	1
32	1	1	1	1	1

D = AB E = AC

Exp.	A	B	C	D	E
1	-1	-1	-1	1	1
2	1	-1	-1	-1	-1
3	-1	1	-1	-1	1
4	1	1	-1	1	-1
5	-1	-1	1	1	-1
6	1	-1	1	-1	1
7	-1	1	1	-1	-1
8	1	1	1	1	1

Confusion

Exp.	A	B	C	D	E	AB	AC	AD	AE	BC	BD	BE	CD	CE	DE
1	-1	-1	-1	1	1	1	1	-1	-1	1	-1	-1	-1	-1	1
2	1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1	1	1	1
3	-1	1	-1	-1	1	-1	1	1	-1	-1	-1	1	1	-1	-1
4	1	1	-1	1	-1	1	-1	1	-1	-1	1	-1	-1	1	-1
5	-1	-1	1	1	-1	1	-1	-1	1	-1	-1	1	1	-1	-1
6	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1	-1	1	-1
7	-1	1	1	-1	-1	-1	-1	1	1	1	-1	-1	-1	-1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Aliases

The following main effects/interactions are aliased:

- $A = BD \leftrightarrow D = AB \leftrightarrow DB = AB^2 = A$
- $A = CE \leftrightarrow E = AC \leftrightarrow CE = AC^2 = A$
- $B = AD$
- $C = AE$
- $D = AB$
- $E = AC$
- $BC = DE$
- $BE = CD$

Resolution

- There are **tables** to know the resolution of a design with factors with two levels based on the number of experiments (rows) and number of factors (columns).
- **Resolution** is generally presented with Roman numerals

	Factores													
Corr	2	3	4	5	6	7	8	9	10	11	12	13	14	15
4	Com	III												
8		Com	IV	III	III	III								
16			Com	V	IV	IV	IV	III	III	III	III	III	III	III
32				Com	VI	IV	IV	IV	IV	IV	IV	IV	IV	IV
64					Com	VII	V	IV	IV	IV	IV	IV	IV	IV
128						Com	VIII	VI	V	V	IV	IV	IV	IV

Pros and cons of Fractional Factorial Designs (FD)

Pros of FD

- **Resource Efficiency**
  - Requires fewer runs than a full factorial design.
- **Focus on Key Effects**
  - Prioritizes main effects and low-order interactions.
- **Scalability**
  - Suitable for experiments with many factors.

Cons of FD

- **Confounding Effects**
  - Higher-order interactions may be aliased with main effects or low-order interactions.
- **Limited Resolution**
  - Designs with low resolution may not fully separate effects of interest.
- **Requires Assumptions**
  - Assumes higher-order interactions are negligible.

Plot the effects

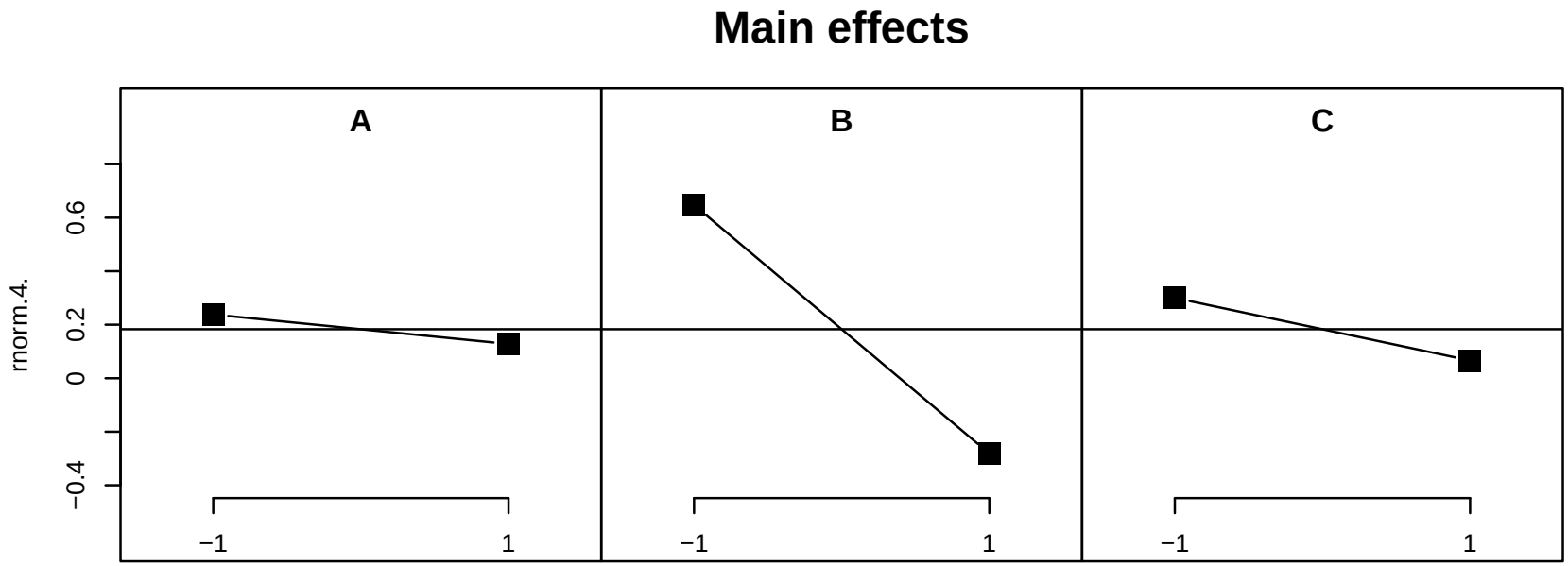
Example

Use the FrF2 function from the FrF2 package to generate a fractional factorial design

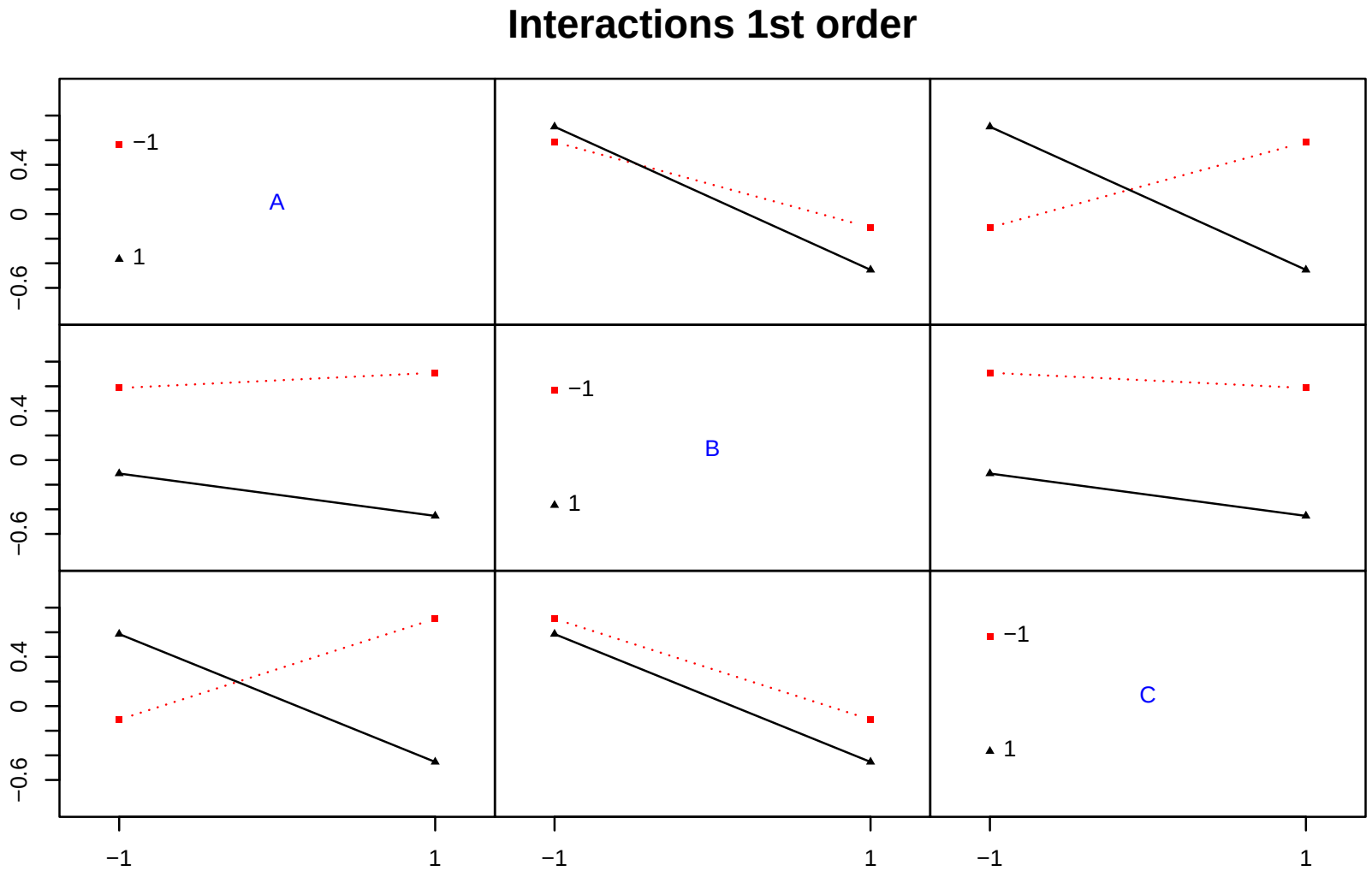
```
library(FrF2)
# Generate a fractional factorial design with 3 factors and 4 runs
fractional_design <- FrF2(nruns = 4, nfactors = 3,
                          factor.names = c("A", "B", "C"))
fractional_design
```

```
##    A  B  C
## 1 -1 -1  1
## 2  1 -1 -1
## 3 -1  1 -1
## 4  1  1  1
## class=design, type= FrF2
```

```r
# Plot main effects
set.seed(12345)
plan <- add.response(fractional_design, response = rnorm(4))
MEPlot(plan, main = "Main effects")
```



```
IAPlot(plan, main = "Interactions 1st order")
```



Other Experimental Designs

Why Explore Other Designs?

Beyond full and fractional factorial designs, specialized designs address unique experimental challenges:

- **Blocking Factors**: Controlling variability due to nuisance factors.
- **High Dimensionality**: Efficient designs for a large number of factors.
- **Non-Factorial Structures**: Handling irregular or constrained factor levels.

Latin Square Designs

Full Factorial Design

- The FFD requires K^p experiments (i.e., combinations).
- **Example**: Bleach Testing
  - Factors
    - **Brand:** $B1, B2, B3$
    - **Conditioning:** Classic, Micro, Liquid
    - **Price:** Cheap, Medium, Expensive
  - Full factorial design: $3^3 = 27$ combinations.

Latin Square Design (LS)

- The full design can be reduced to **9** combinations using a **Latin Square**.
- Internal values are obtained by shifting the columns (or rows) one position upwards (or to the left).

Type/Brand	B1	B2	B3
Classic	Cheap	Medium	Expensive
Micro	Medium	Expensive	Cheap
Liquid	Expensive	Cheap	Medium

Latin Square Designs with nuisance factors

Latin Square with nuisance factors

- **Purpose**: LS could be used to reduce variability when several nuisance factors need to be controlled.
- **Structure**: A square matrix where each row and column represents a blocking factor, and treatments are assigned once per row and column.
- **Example**:

A	B	C
B	C	A
C	A	B

Each treatment (A, B, C) appears exactly once in each row and column.

Example

Use the `DoE.base` R package to generate a Latin Square design:

```
# Create a 3x3 Latin Square design
latin_square <- oa.design(nruns = 9, nfactors = 3, nlevels = 3,
                          factor.names = c("Row", "Column", "Treatment"))
latin_square
```

```
##   Row Column Treatment
## 1   3      2         1
## 2   1      2         3
## 3   2      3         1
## 4   1      3         2
## 5   3      3         3
## 6   1      1         1
## 7   3      1         2
## 8   2      2         2
## 9   2      1         3
## class=design, type= oa
```

Optimal Designs

Optimal Designs

- **Purpose**: Find the most efficient design for a given experimental setup.
- Minimizes the average variance of parameter estimates.
- Useful for experiments with constraints or **irregular factor levels**.

Variance in Linear Models

The variance of the estimates in a linear model is given by:

$$V(\mathbf{b}) = \sigma^2 \cdot (\mathbf{X}'\mathbf{X})^{-1} \quad (\mathbf{X}' \text{ is the transpose of } \mathbf{X})$$

- Since σ^2 is constant, the goal is to minimize:

$$(\mathbf{X}'\mathbf{X})^{-1}$$

- Which implies maximizing:

$$(\mathbf{X}'\mathbf{X})$$

- In other words, seek the maximum determinant.

Example

Use the optFederov function from the AlgDesign R package to create a D-Optimal Design

```
library(AlgDesign)
# Define factors and levels
factors <- data.frame(Factor1 = rep(c("Low", "Medium", "High"),
                                    each = 2),
                      Factor2 = c("A", "B", "A", "B","A", "B"))
# Generate a D-optimal design
d_optimal <- optFederov(data = factors, nTrials = 4)
d_optimal$design
```

```
##   Factor1 Factor2
## 1     Low       A
## 2     Low       B
## 3  Medium       A
## 5    High       A
```

Rules of thumbs for choosing the best design

- Choosing the right design depends on the problem's constraints and goals.

Select the design

- **FFD**: No budget constraint.
- **FD**: Easy reduction of FFD.
- **LS Designs**: Effective for controlling two nuisance factors.
- **Optimal Designs**: Tailored for irregular experimental setups to maximize efficiency.

Statistical Analysis in DOE

What is Statistical Importance?

Statistical importance in DOE quantifies the significance of factors and their interactions:

- Determines whether changes in factors lead to meaningful variations in the response.
- Helps prioritize which factors have the most impact on the outcome.

How is Statistical Importance Assessed?

- **ANOVA (Analysis of Variance)**:
  - Tests the null hypothesis that a factor has no effect on the response.
  - Provides `p-values` and `F-statistics` to evaluate factor significance.
- **Effect Size**:
  - Measures the magnitude of a factor's influence on the response.
  - Example: Partial η^2 (proportion of variance explained).
- **Pareto Charts**:
  - Visualize factor effects ordered by importance.

ANOVA

- **Problem**: Evaluate the impact of `batch size` and `learning rate` on `model accuracy`.
- **Solution**:
  - Significant factors ($p < 0.05$) indicate important effects.
  - Interaction terms reveal combined effects.

```r
anova_res <- aov(accuracy ~ batch_size * learning_rate, data = data)
summary(anova_res)
```

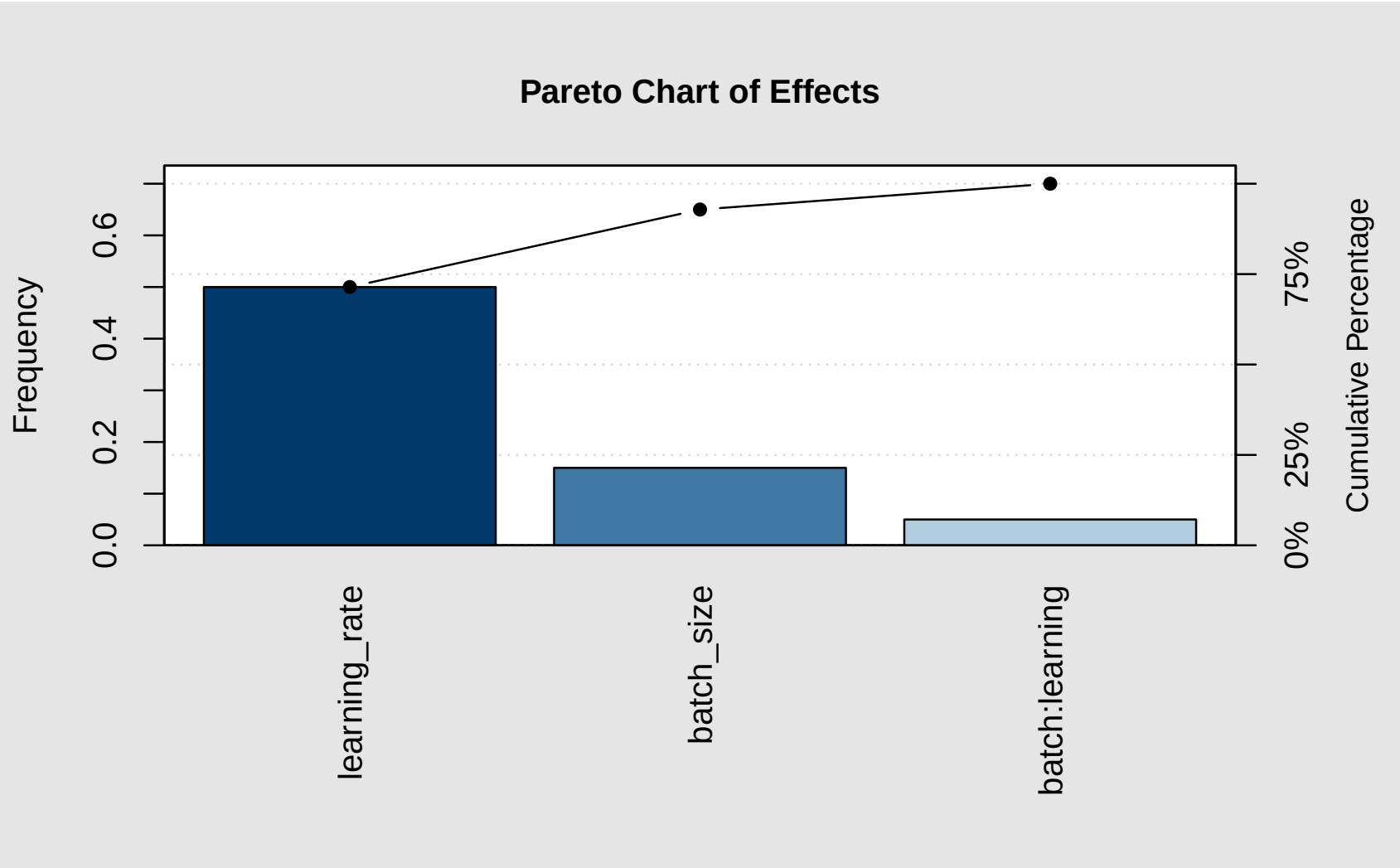
```
##                              Df    Sum Sq  Mean Sq F value  Pr(>F)
## batch_size                    1 0.000579 0.000579   0.555 0.48994
## learning_rate                 1 0.028605 0.028605  27.421 0.00336 **
## batch_size:learning_rate      1 0.000000 0.000000   0.000 1.00000
## Residuals                     5 0.005216 0.001043
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualizing Statistical Importance

- **Pareto Charts**:
  - Highlight the relative importance of factors.
  - Example in R using qcc package:
- **Main Effects Plots**:
  - Show the direct impact of each factor on the response.

```r
library(qcc)
model <- lm(accuracy ~ batch_size * learning_rate, data = data)
effects <- abs(coef(model))[-1]
```

```r
pareto.chart(effects, main  = "Pareto Chart of Effects")
```



References

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2. Box, G. E. P., Hunter, W. G., & Hunter, J. S. (2005). Statistics for experimenters: Design, innovation, and discovery (2nd ed.). Wiley-Interscience. ISBN: 978-0471703136